

INTERNATIONAL STANDARD

**Environmental testing –
Part 3-5: Supporting documentation and guidance – Confirmation of the
performance of temperature chambers**



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INTERNATIONAL STANDARD

**Environmental testing –
Part 3-5: Supporting documentation and guidance – Confirmation of the
performance of temperature chambers**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

ENVIRONMENTAL TESTING –

**Part 3-5: Supporting documentation and guidance –
Confirmation of the performance of temperature chambers**

FOREWORD

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International Standard IEC 60068-3-5 has been prepared by IEC technical committee 104: Environmental conditions, classification and methods of test.

This second edition cancels and replaces the first edition published in 2001. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) Confirmation procedures are clarified.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
104/759/FDIS	104/778/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 60068 series, published under the general title *Environmental testing*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

IEC 60068 (all parts) contains fundamental information on environmental testing procedures and severities.

The expression "environmental conditioning" or "environmental testing" covers the natural and artificial environments to which components or equipment may be exposed so that an assessment can be made of their performance under conditions of use, transport and storage to which they may be exposed in practice.

Temperature chambers used for "environmental conditioning" or "environmental testing" are not described in any publication, although the method of maintaining and measuring temperature and/or humidity has a great influence on test results. The physical characteristics of temperature chambers can also influence test results.

ENVIRONMENTAL TESTING –

Part 3-5: Supporting documentation and guidance – Confirmation of the performance of temperature chambers

1 Scope

This part of IEC 60068 provides a uniform and reproducible method of confirming that temperature test chambers, without specimens, conform to the requirements specified in climatic test procedures of IEC 60068-2 (all parts) and other standards. This document is intended for users when conducting regular chamber performance monitoring.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60068-2 (all parts), *Environmental testing – Part 2: Tests*

IEC 60068-3-7, *Environmental testing – Part 3-7: Supporting documentation and guidance – Measurements in temperature chambers for tests A and B (with load)*

IEC 60068-3-11, *Environmental testing – Part 3-11: Supporting documentation and guidance – Calculation of uncertainty of conditions in climatic test chambers*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

temperature test chamber

enclosure or space in some parts of which the temperature conditions, specified in IEC 60068-2 (all parts), can be achieved

3.2

temperature setpoint

desired temperature as set by the chamber controls

3.3

achieved temperature

stabilized temperature which desired temperature at the centre of the working space achieves within specified tolerance

3.4**temperature stabilization**

state of maintaining temperature within specified tolerance during specified time at specified points in the working space

3.5**temperature fluctuation**

difference, after stabilization, between the maximum and minimum temperatures at specified point in the working space during a specified interval of time

Note 1 to entry: For calibration, the centre point of working space may be used.

3.6**working space**

part of the chamber in which the specified conditions can be maintained within the specified tolerances

Note 1 to entry: See Figure 1 and Table 1.

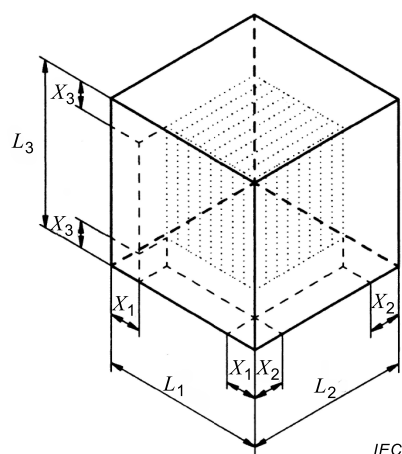


Figure 1 – Working space

Table 1 – Practical dimensions

Size	Volume l	Distance X mm	X (min.) mm
Small	Up to 1 000	$L/10$	50
Medium	1 000 to 2 000	$L/10$	100
Large	More than 2 000	$L/10$	150
NOTE Not all chambers are cubic in construction.			

3.7**temperature gradient**

maximum difference in mean value, after stabilization, at any moment in time between two separate points in the working space

3.8**temperature variation in space**

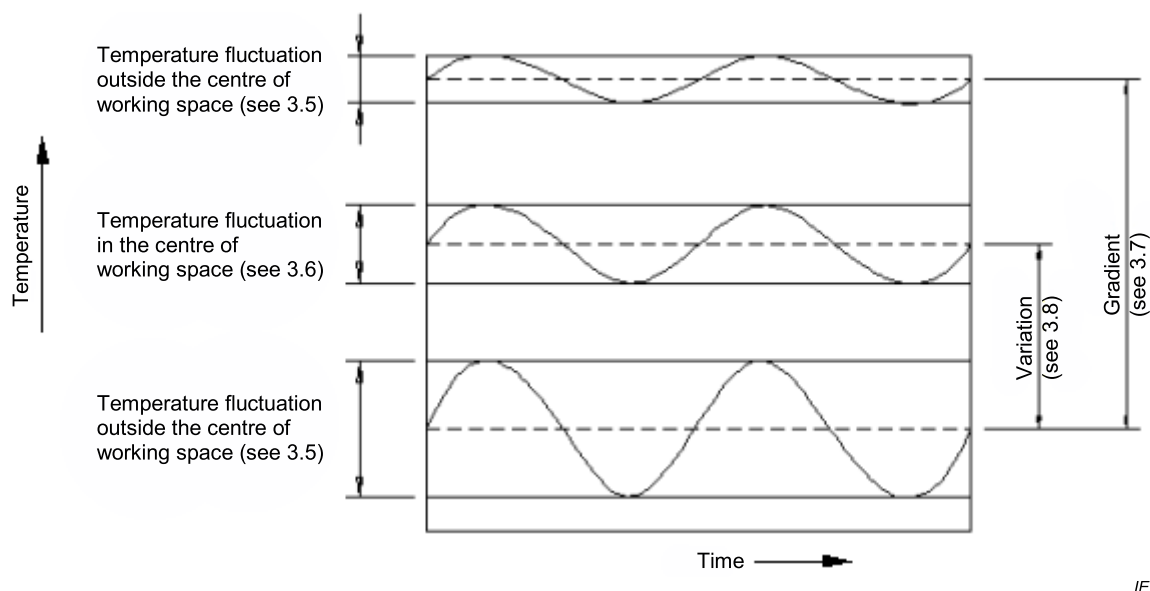
difference in mean value, after stabilization, at any moment in time between the temperature at the centre of the working space and at any other point in the working space

3.9

temperature rate of change

rate, in kelvin per minute, for the transition between two specified temperatures measured at the centre of the working space

Note 1 to entry: See Figure 2.



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Figure 2 – Example of temperature differences

4 Measuring chamber performances

4.1 Test area environment

The environment around a temperature test chamber may influence the conditions inside the test chamber.

The confirmation of performance of temperature chambers should be carried out under standard atmospheric conditions specified in IEC 60068-1.

4.2 Temperature measurement system

The uncertainty of measurement of the output of the measurement system should be determined by calibration of the system, traceable to international standards (see ISO 10012).

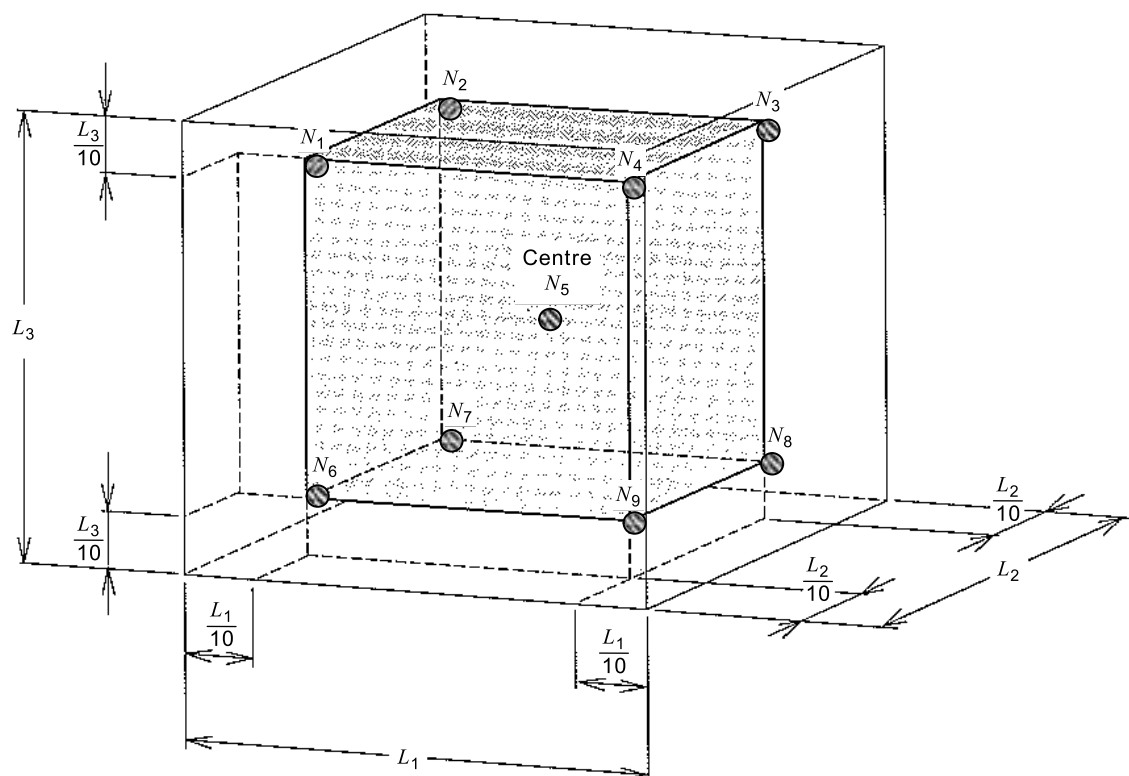
The temperature sensors may be either calibrated platinum resistors or a thermocouple. The thermal response time of the sensors shall be within a minimum of 10 s and a maximum of 40 s for 50 % of response. It is preferred that the thermal response time of the entire measurement system to be less than 40 s. The use of sensors that are compliant to IEC 60584-1 tolerance class 1 (for thermocouples) or IEC 60751 tolerance class A (for resistors) is recommended.

4.3 Temperature chamber test specimens

All measurements described in 4.5 are performed with an empty working space. For measuring with test specimens (with or without heat dissipation), see IEC 60068-3-7.

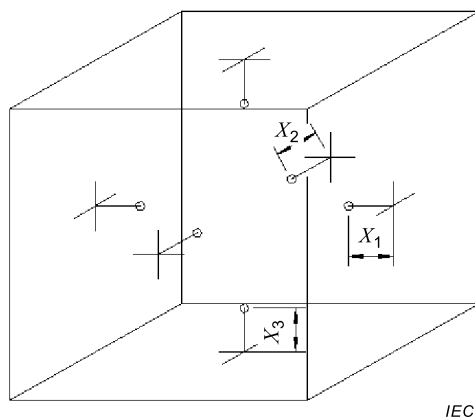
4.4 Specified location of temperature sensors in working space

Temperature measuring sensors are located in each corner and in the centre of the working space (see Figure 3, minimum 9 sensors). For temperature chambers over 2 000 l, additional sensors should be located in front of the centre of each wall (see Figure 4, minimum 15 sensors). The measuring system is to be arranged in such a way that the temperature distribution of the empty test chamber will not be affected. For a large capacity chamber, there may be a significant difference between the temperature control sensor(s) and the temperature at the centre of the working space. It may be necessary to adjust the temperature setting to achieve the necessary tolerance.



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Figure 3 – Location of sensors for temperature chambers up to 2 000 l



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Figure 4 – Location of minimal additional sensors for temperature chambers over 2 000 l

For confirmation monitoring, data should be recorded at least once a minute. The device used for recording data from the chamber monitoring sensors should be independent of the chamber control system.

4.5 Measurement method

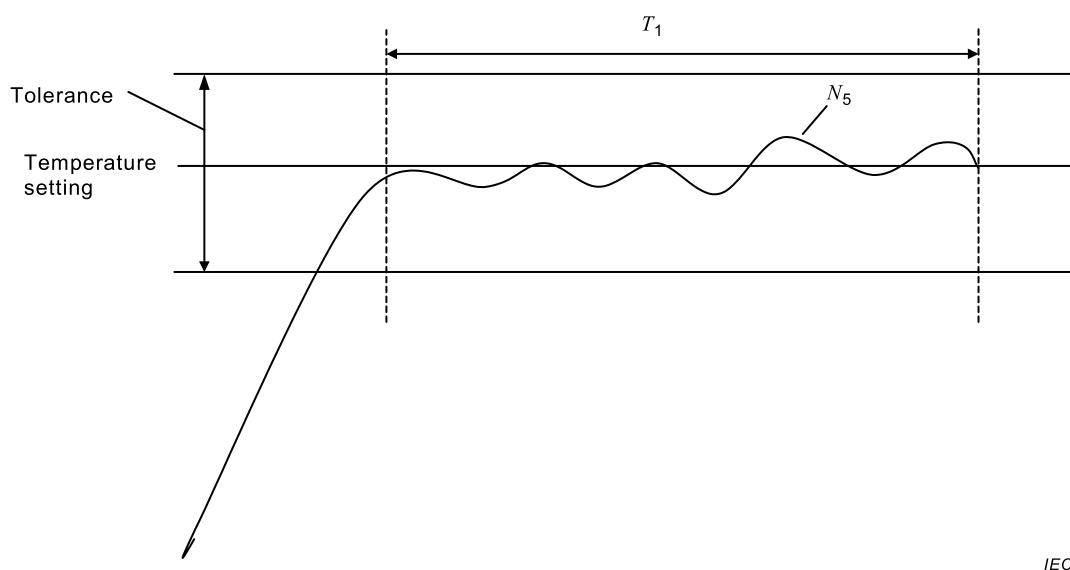
4.5.1 General

The temperature output of the temperature measuring system (see Figure 3 or Figure 4) determines, after chamber stabilization, the achieved temperature, temperature fluctuation and temperature gradient of the working space. For tolerance, the specification of the temperature/humidity chamber or, as necessary, tolerance specified in IEC 60068-2 (all parts), is required to maintain at the centre of the working space. Location of sensor is minimum 9 points or 15 points. This depends on the test chamber size. The measurement method is explained based on 9 points.

Uncertainty of measurement of the temperature measuring system shall be according to IEC 60068-3-11.

4.5.2 Achieved temperature

Temperature is achieved when the centre of the working space maintains the tolerance as required by IEC 60068-2 (all parts). An example is shown in Figure 5.



For tolerance, check the specification of the temperature chamber, or, as necessary, use tolerance specified in IEC 60068-2 which is required to maintain at the centre of the working space.

T_1 must be minimum 30 min. N_5 is the temperature at the centre of the working space.

Figure 5 – Example of achieved temperature

4.5.3 Temperature stabilization

Temperature reached and maintained within the allowable range in the working space is shown in Figure 6. Allowable range is based on the temperature fluctuation, temperature variation in space, and temperature gradient as the temperature chamber specification. Specified time T_2 is minimum 30 min after the measurement points (e.g. N_1 to N_9) are within the allowable range.

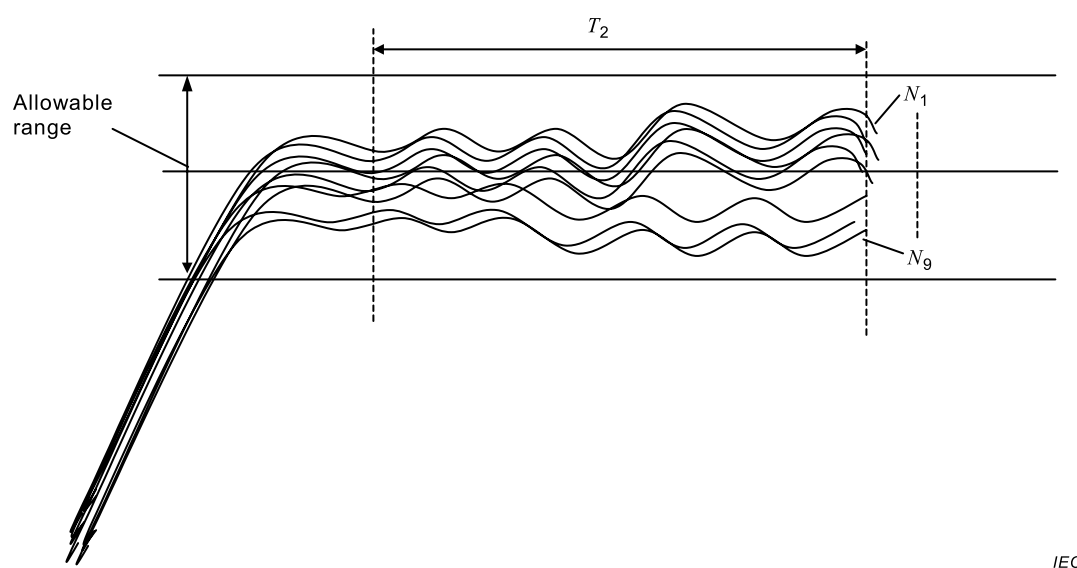


Figure 6 – Example of temperature stabilization for chambers up to 2 000 l

4.5.4 Temperature fluctuation

The fluctuation during a specified interval of time at specified temperature points in the working space, after temperature stabilization, is shown in Figure 7.

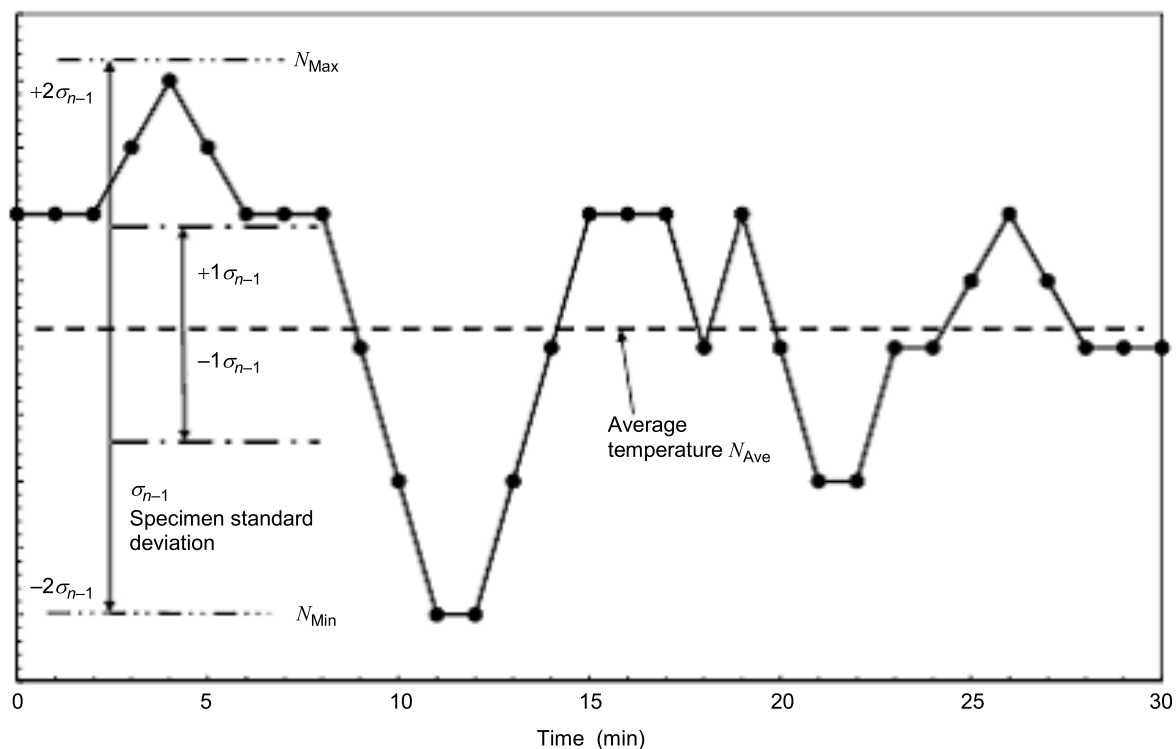


Figure 7 – Example of temperature fluctuation

Measurements are actually made at a certain sampling interval. It is not guaranteed that data captures the fluctuation peak. For that reason, the sample standard deviation, σ_{n-1} shall be obtained from the data measured in each measurement point after temperature has stabilized, and the temperature fluctuation shall be within $\pm 2\sigma_{n-1}$.

The temperatures measured in a certain measurement point, after stabilization, over a 30-minute period 10 or more times at even intervals shall be as follows.

$$N_1, N_2, \dots, N_i, \dots, N_n \quad (n \geq 10)$$

The mean of N shall be N_{Ave} .

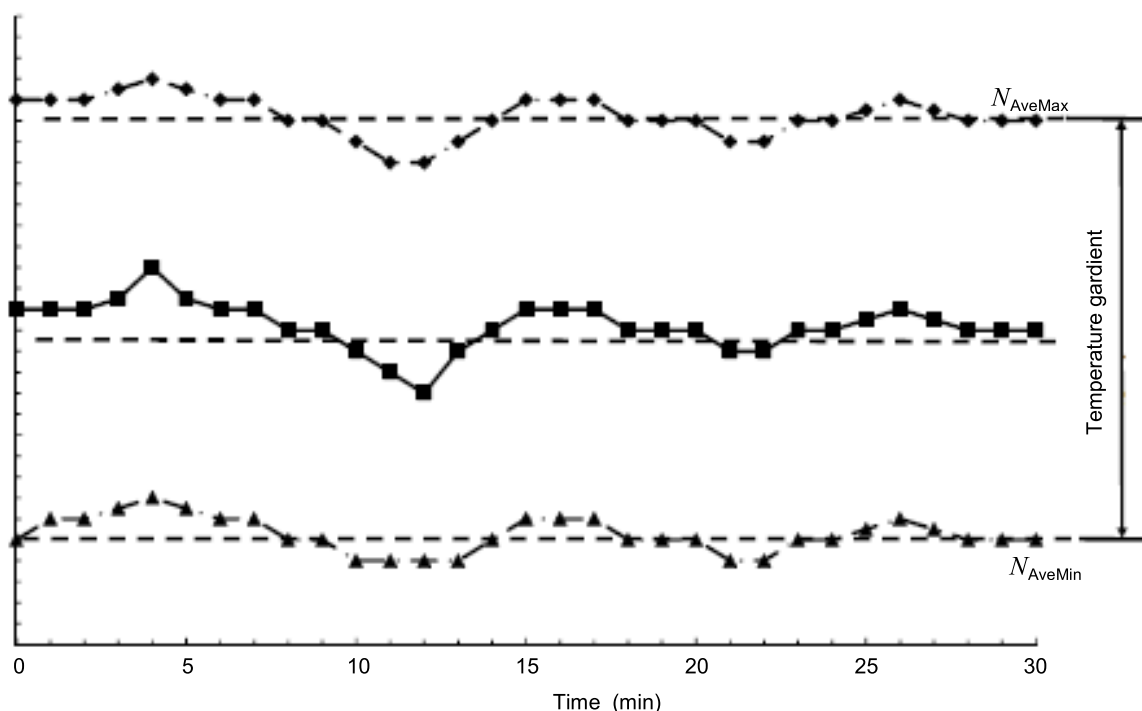
The specimen standard deviation, σ_{n-1} shall be defined as follows.

$$\sigma_{n-1} = \sqrt{\frac{\sum (X_i - N_{Ave})^2}{n - 1}}$$

Temperature fluctuation $\pm 2\sigma_{n-1}$ shall be obtained in all 9 measurement points, and the highest value within that shall be annotated as the temperature fluctuation.

4.5.5 Temperature gradient

As shown in Figure 8, the maximum difference in mean temperature in all measurement points of the effective space shall be the temperature gradient.



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N_{AveMax} : Mean highest temperature in each of 9 measurement points

N_{AveMin} : Mean lowest temperature in each of 9 measurement points

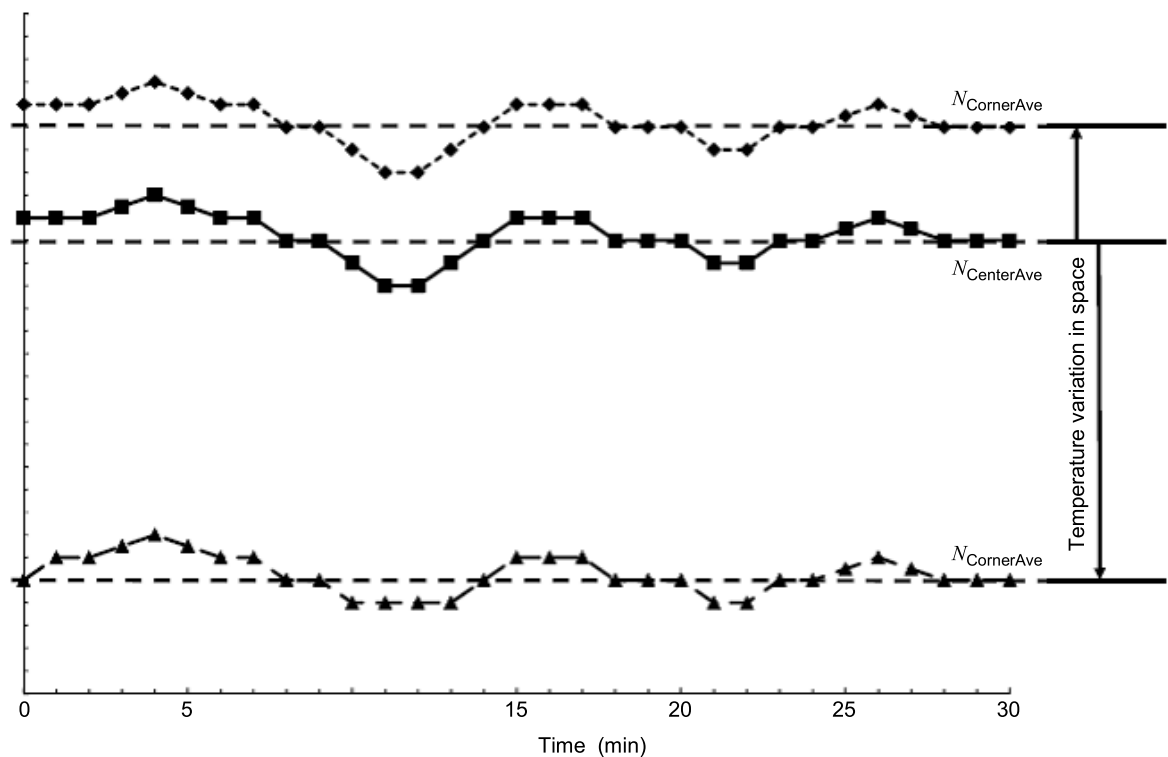
Temperature gradient = $N_{AveMax} - N_{AveMin}$

Figure 8 – Example of temperature gradient for chambers up to 2 000 l

4.5.6 Temperature variation in space

As shown in Figure 9, temperature variation in space is the difference of the mean temperature at the centre of the working space and the mean temperature of another

measurement point. The maximum difference between the centre of the working space and each measuring point shall be stated.



IEC

$N_{CenterAve}$: Mean temperature at chamber center

$N_{CornerAve(j)}$ $j=1$ to 8: Mean temperature at corner of working space

Temperature variation in space = $|\text{Max}(N_{CornerAve[j]} - N_{CenterAve})|$

The value obtained as the temperature variation in space shall be $(N_{CornerAve(j)} - N_{CenterAve})$, and the absolute value of the highest difference obtained for the 8 corners shall be annotated as the temperature variation in space

Figure 9 – Example of temperature gradient for chambers <2 000 L

4.5.7 Temperature rate of change

As shown in Figure 10, temperature rate of change between specified temperatures shall be calculated with the following method, and indicated in K/min.

Temperature heat-up rate = $\Delta t/T_1$

Temperature cool-down rate = $\Delta t/T_2$

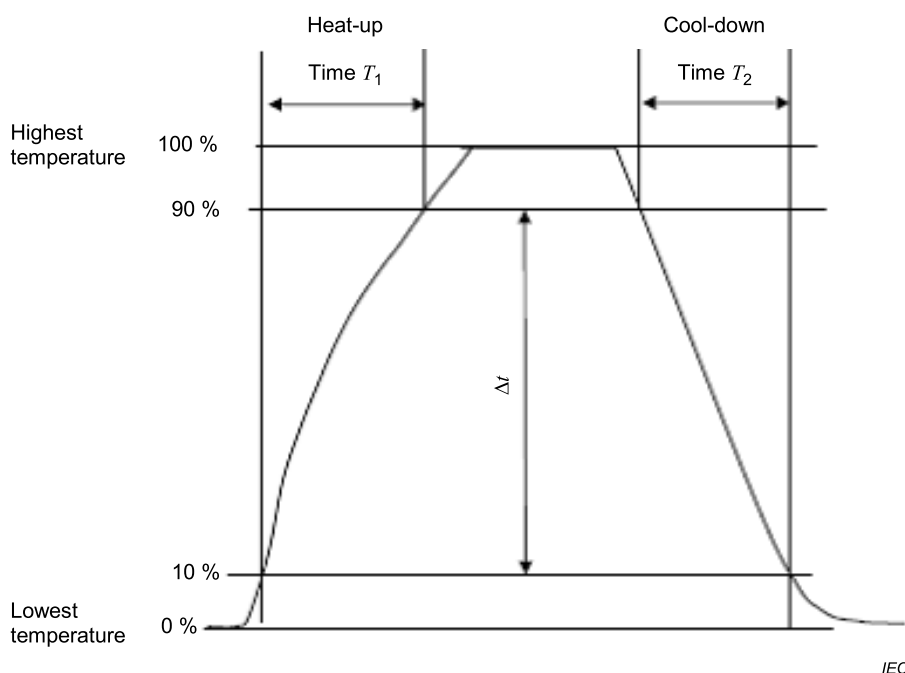


Figure 10 – Example of temperature rate of change

To measure temperature rate of change:

- adjust chamber to lowest specified temperature and allow to stabilize;
- adjust chamber to highest specified temperature, monitoring the time between the 10 % and the 90 % points of the temperature range;
- allow chamber to stabilize at the highest specified temperature;
- adjust chamber to lowest specified temperature, monitoring the time between the 90 % and 10 % points of temperature range.

4.6 Standard temperature sequence

The following test sequence is considered to be the minimum recommended to obtain the necessary data for confirmation of the performance of a temperature chamber.

The test sequence is as follows:

- start at ambient conditions;
- adjust chamber to highest specified temperature and allow chamber to stabilize;
- measure performance at highest temperature;
- adjust chamber to lowest specified temperature, monitoring rate of change and allow chamber to stabilize;
- measure performance at lowest temperature;
- adjust chamber to highest specified temperature, monitoring rate of change;
- adjust chamber to atmospheric conditions and allow chamber to stabilize;
- measure performance at atmospheric conditions.

5 Information to be given in the performance test report

As a minimum, the following information shall be contained in the test report.

- a) Atmospheric conditions in the test area for measurement.
- b) Size and volume of chamber enclosure and working space.
- c) Temperature fluctuation, temperature variation in space and temperature gradient.
- d) Temperature rate of change.
- e) Highest/Lowest temperature.
- f) Measurement results from each measurement position.
- g) Details of data acquisition systems.
- h) Measurements uncertainties.
- i) Fixtures used for measurement.

Bibliography

IEC 60068-1, *Environmental testing – Part 1: General and guidance*

IEC 60068-3-6, *Environmental testing – Part 3-6: Supporting documentation and guidance – Confirmation of the performance of temperature/humidity chambers*

IEC 60584-1, *Thermocouples – Part 1: EMF specifications and tolerances*

IEC 60751, *Industrial platinum resistance thermometers and platinum temperature sensors*

ISO 10012, *Measurement management systems – Requirements for measurement processes and measuring equipment*

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